

Physik II — FS 2024 — Prof. R. Wallny — Serie 6

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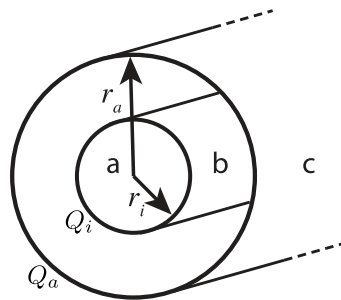
Exercise 1: Two conducting spheres

Objective: Considerations on initial and final conditions, conservation of charge, application of the Gauss's law.

The centers of two conducting spheres with radii R_1 and R_2 are at a distance $d \gg R_1, R_2$ from each other. The spheres carry the charges Q_1 and Q_2 , respectively. Now, they will be connected by an infinitely thin conducting wire. Compute the charges on the spheres, their electrostatic potentials and the electric fields at their respective surface after the connection.

Exercise 2: Cylindrical capacitor

Objective: Calculate capacitances.



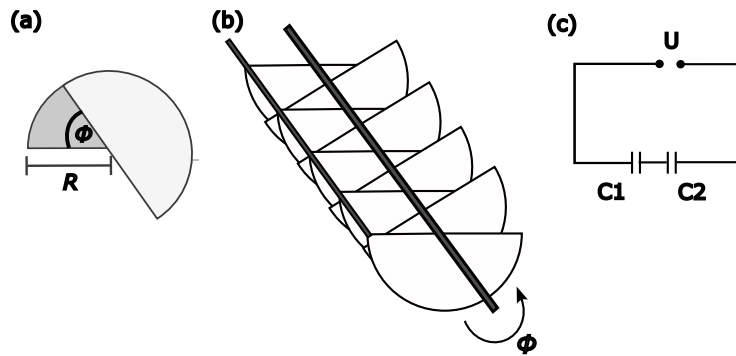
Consider a cylindrical capacitor of length h with inner and outer radii r_i and r_a . The respective charges on the cylinders are Q_i and Q_a .

- Calculate the electric field in the regions a , b , and c .
- Consider now that each cylinder has two surfaces (inner and outer). Determine the charge that is held on each of these four surfaces.
- Calculate the capacitance of this capacitor.

Exercise 3: Rotary Variable Capacitor

Objective: Capacitor circuits.

A rotary variable capacitor is a mechanically variable capacitor that consists of a shaft and same size semicircular capacitor plates of radius R which are arranged in parallel with a distance d . Every second plate is rotatable with rotation angle ϕ .

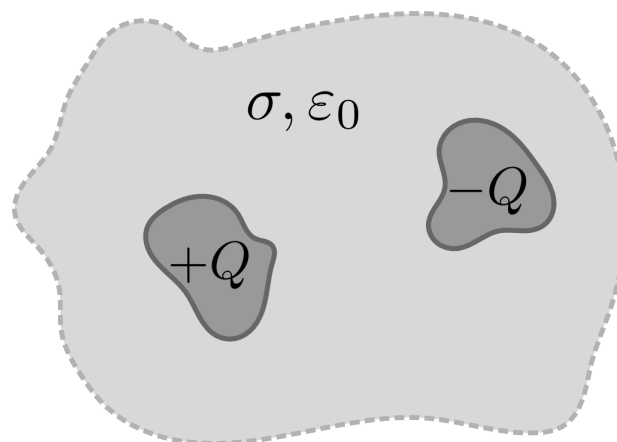


- a) Consider two semicircular capacitor plates with distance d which are rotated against each other by an angle ϕ (see (a) in the figure). Determine the capacitance of this capacitor as a function of ϕ .
- b) The setup in figure (b) consists of 5 static capacitor plates and 4 rotatable capacitor plates. Sketch the circuit diagram of the setup. Calculate the maximum capacitance of this capacitor with $R = 3 \text{ cm}$ and $d = 0.05 \text{ mm}$. How much energy can be stored in the fully charged capacitor if it is charged from a voltage source with $V = 3 \text{ V}$? Neglect the recess areas for the shaft.
Hint: In this case, the approximation $d \ll R$ is valid.
- c) In figure (c) two of the capacitors shown in (b) are connected in series. Both capacitors are connected to the same voltage. Calculate the maximum capacitance of this setup.

Exercise 4: Does the shape play a role?

Objective: Gaussian law, Ohm's law, currents.

Two metallic electrodes of unknown shape carry the charges $+Q$ and $-Q$ respectively and are embedded into an infinitely extended material with (ohmic) conductivity σ and dielectric constant ε_0 (i.e. the dielectric constant of the vacuum).



- a) Let C be the capacitance of the system described above. Show that the resistance R between the electrodes can be expressed as

$$R = \frac{\varepsilon_0}{\sigma C}. \quad (1)$$

Hint: To show this, consider first the current, that flows out of one of the electrodes and write it as the flow of current through a closed surface that completely encloses the electrode.

- b) The resistance in equation (1) seems to be independent of the shape and geometry of the system. Is this really the case? Briefly explain your answer.
- c) Assume, that at time $t = 0$ the electrodes are charged with $+Q_0$ and $-Q_0$ respectively. The electrodes now discharge with the current calculated above. Show that the temporal development of the charges for $t > 0$ has the following shape:

$$Q(t) = Q_0 e^{-t/\tau}, \quad (2)$$

with

$$\tau = \frac{\varepsilon_0}{\sigma}. \quad (3)$$

Exercise 5: Current through cube

Objective: continuity equation, current density, current through surface.

We have the following charge density

$$\vec{J} = \begin{pmatrix} x^2 \\ -2y^2 \\ 3z \end{pmatrix} \text{ mA cm}^{-2} \quad (4)$$

with the local coordinates x , y and z in centimetres.

- a) Determine the temporal development of the charge density $\frac{d\rho}{dt}$.
- b) Determine the current $I_{\partial V}$ through the surface of the cube $-1 \text{ cm} \leq x, y, z \leq 1 \text{ cm}$.
- c) The cube is uncharged at $t = 0$ s. What charge does it have after $\Delta t = 10$ s?